Varadraj Patil

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Education

Vishwakarma Institute of Technology — Pune, Maharashtra
Bachelor of Technology in Computer Engineering (Software Engineering) (CGPA: 8.96/10) — *Aug. 2024 – May 2028*

Projects

Distributed Peer-to-Peer Storage Network | *Go, gRPC, React, Kademlia DHT, Protocol Buffers* | GitHub

- Built a decentralized content-addressed storage network using the Kademlia DHT with O(log N) peer discovery and SHA-256 chunking for efficient content retrieval.
- Implemented Merkle DAGs, 3-way replication, gRPC communication, and self-healing heartbeat monitoring to provide fault-tolerant storage across 20 scattered nodes.

Bank Transaction System | *Node.js, Express, MongoDB, Mongoose, JWT* | GitHub

- Developed a double-entry ledger-based payment processing system with ACID-compliant transactions, JWT authentication, and role-based authorization for secure financial operations.
- Engineered idempotent fund transfers, immutable audit logs, and token blacklisting, ensuring transaction consistency, fraud prevention, and reliable recovery from duplicate requests.

Behavioral Biometric Authentication System | *Python, Scikit-learn, Tkinter, Pandas* | GitHub

- Developed a behavioral biometric authentication system using mouse dynamics collected from 4 custom mini-games, extracting 40+ behavioral features across multiple gameplay sessions.
- Trained Random Forest and SVM classifiers, achieving 88.5% authentication accuracy while supporting user registration, model training, and real-time authentication.
- Built a desktop application with Tkinter supporting user registration, model training, authentication, evaluation, and feature visualization.

Academic Projects

Automated DSA Tracking Pipeline | GitHub — July 2026
Vishwakarma Institute of Technology

- Built a scheduled GitHub Actions automation pipeline that integrates external APIs, processes solved problems, and automatically commits structured README updates.

Real-Time Messaging Platform | GitHub — Feb. 2026
Vishwakarma Institute of Technology

- Designed a real-time 1-to-1 messaging platform using the MERN stack and Socket.IO with JWT authentication, delivering instant messaging, live presence tracking, and secure WebSocket communication.

Certifications

Microsoft & LinkedIn: Career Essentials in Generative AI — Google: AI Essentials — Google: Prompting Essentials — IEEE ESCI 2026: Research Paper Presentation

Technical Skills

Languages: C/C++, Go, Python, Java, JavaScript, SQL, HTML/CSS
Core Concepts: Data Structures & Algorithms, System Design, Object-Oriented Programming, Operating Systems, DBMS, COA, Distributed Systems, Computer Networks, Mainframe Computing, PLM
Technologies: Node.js, Express.js, REST APIs, React, Next.js, Redis, OAuth 2.0, Ngrok
DevOps/Cloud: AWS EC2, GitHub Actions, CI/CD, Vercel, Render, Docker, Linux (Arch, Ubuntu), Git, GitHub
Database: MongoDB, MySQL, Supabase
AI/ML: Scikit-learn, Pandas, NumPy, Random Forest, SVM